

Anirudh Phophalia

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EDUCATION

Thapar Institute of Engineering and Technology

B.Tech in Computer Science and Engineering, CGPA: 9.61

Indian Institute of Technology Madras

B.S. in Data Science and Applications

Patiala, India

Aug 2024 – Present

Remote

May 2024 – Present

EXPERIENCE

Samsung R&D Institute Bangalore

Aug 2026 – Present

Samsung PRISM Project Intern

Remote

- Developing an **AI-driven Smart Data Privacy Scanner** using **Multimodal Computer Vision, OCR, and Semantic Analysis** to detect Personally Identifiable Information (PII), such as faces, ID documents, signatures, and text-based identifiers.

Defence Research and Development Organisation (DRDO)

Jun 2026 – Present

Summer Research Intern

Remote

- Developing **SDR-based drone detection** and **RF signal analysis** pipelines using signal processing and machine learning techniques for drone communication identification.

Cambiante Technologies

Jun 2025 – Jul 2025

Frontend Developer Intern

Jaipur

- Developed core **Angular** modules for **Cam-Shot**, an AI-powered event photo management platform, including authentication, photo galleries, and event management interfaces with a responsive user experience.
- Integrated **REST APIs** and optimized frontend performance by resolving **UI bugs**, improving component reusability, and streamlining data flow.

Backslash Computing Society, TIET

Sep 2024 – Dec 2025

Technical Core Member

Patiala

- Led the end-to-end development of the **Cryptic Hunt Portal**, enabling a smooth experience for 500+ participants during the society's flagship technical event **Synapse**.
- Mentored junior members through workshops on Web Development, Git/GitHub, and AI/ML, fostering collaborative development practices and accelerating technical onboarding.

PROJECTS

Research Publication | Transformer-Based Maritime Semantic Segmentation

Zenodo

- Developed a transformer-based maritime semantic segmentation framework using **PyTorch**, Swin-L, Mask2Former, and the **OpenMMLab** ecosystem (MMCV, MMDetection, MMSegmentation) for autonomous USVs.
- Trained and evaluated on the **LaRS** maritime dataset for water, sky, and obstacle segmentation, achieving a **97.71% F1-score at 12.33 FPS**.
- Published the work as *“Transformer-Driven Semantic Segmentation for Robust Autonomous Inspection in Unmanned Surface Vessels”* (Authors: A. Phophalia, P. Mittal, K. Saini, V. Mehta), currently under review at *The Journal of Supercomputing (Springer)*.

Steel Micro-Crack Detection using ResNet18, U-Net and XAI | PyTorch, FastAPI, ONNX, Streamlit

- Built a two-stage defect inspection pipeline on 12K+ **Severstal steel images**, achieving **99.28% accuracy, 0.988 ROC-AUC, and Dice Score of 0.71**.
- Integrated **Grad-CAM** explainability and deployment support through **TorchScript, ONNX, FastAPI, and Streamlit**.

MedInternia | MongoDB, Express.js, React, Node.js

Indian–Israeli Hackathon

- Developed a **healthcare collaboration platform** enabling doctors and patients to post anonymized clinical cases for interns to analyze, supporting AI-assisted learning through guided discussions and competency-based recommendations.
- Selected among the **Top 50** projects in **GSSoC 2026** with **90+ contributors**; serving as **Project Admin** leading contributor onboarding, code reviews, feature planning, and open-source collaboration workflows.

SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, R, MATLAB

Machine Learning & AI: PyTorch, TensorFlow, Computer Vision, Deep Learning, Transformer Models, OCR

Web Development: Angular, React, Node.js, Express.js, NestJS, Flask

Databases: PostgreSQL, MongoDB, MySQL, OracleDB, SQLite

Tools & Platforms: Git, Docker, Linux, AWS, Postman, VS Code, Jira